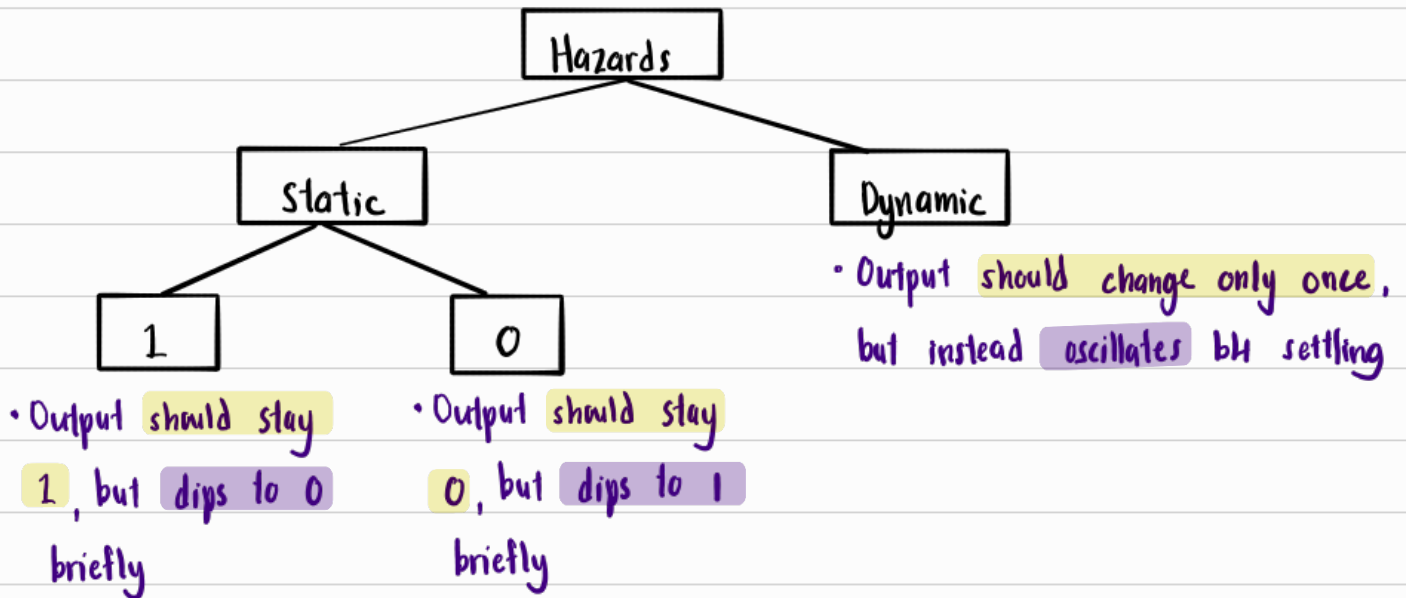
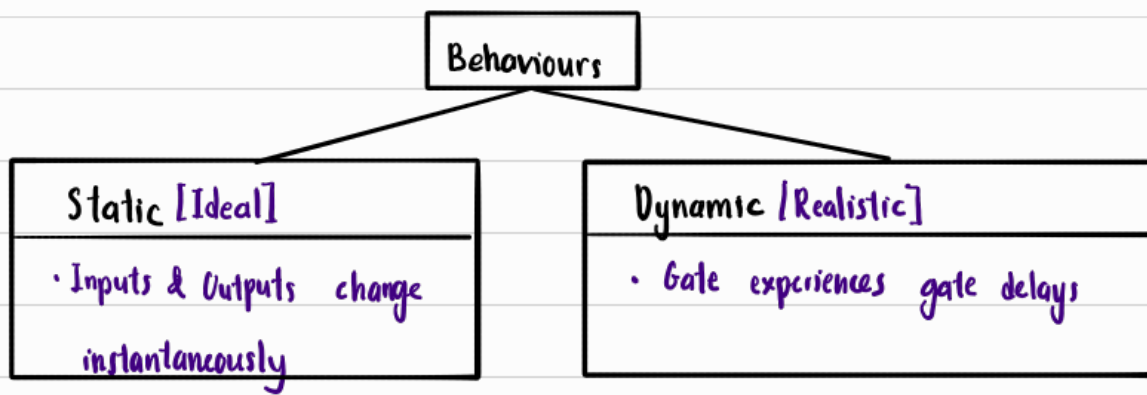
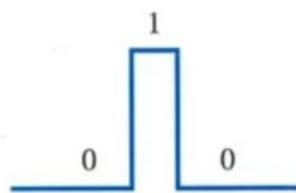


(a) Static 1-hazard



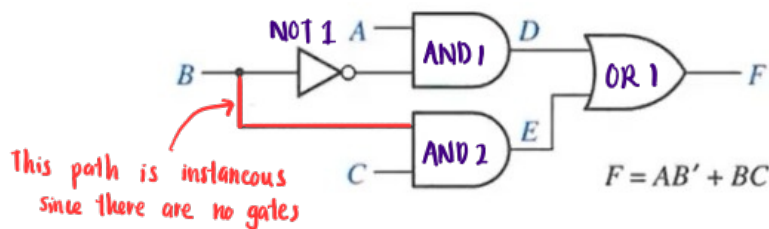
(b) Static 0-hazard



(c) Dynamic hazards

Example 1

Given a circuit with output, $F = AB' + BC$, we observe a static 1-hazard when B changes from 1 to 0, while A and C remain 1.

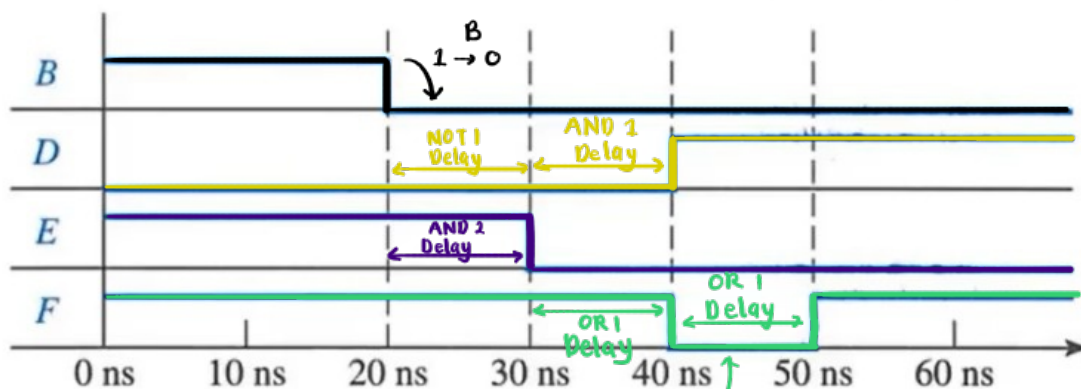


Combination of minterms

A \ BC	0	1
00	0	1
01	0	1
11	1	1
10	0	0

Transition from Implicant 1 to Implicant 2

1-hazard



Output is zero becz at this moment, E have changed to 0 while D have not changed to 1 due to the extra NOT gate
 $\therefore 0 \text{ or } 0 \text{ gives an output of } 0$

The Logic Behind it

$$F = AB' + BC \longrightarrow \text{This is the SOP, or aka. Sum of "Minterms".}$$

True conditions: $A=1, B=0$ $B=1, C=1$

$\therefore F$ is True & only true when AB' or BC is true.

1) When 1 bit changes [Adjacent Row or Columns of K-map],

here only B changes from 1 to 0, BC turns off, AB' turns on.

2) This transition from BC to AB' isn't instant

$\hookrightarrow$ BC drops to 0 first [Bcz AB' path has an extra NOT gate]

$\hookrightarrow$ AB' still 0 because of the extra delay from the extra NOT gate

$\therefore$ Both terms 0, output $[0 + 0] = 0$ [Static 1 hazard]

So to solve this hazard, we have to introduce a "bridge term" that bridges that time delay. This bridge term have to be independent of the changing variable [B]

		A	
		0	1
BC	00	0	1
	01	0	1
	11	1	1
	10	0	0

This product term will be independent of B
 $\therefore AC$

1-hazard

$$\therefore F_{\text{no hazard}} = AB' + BC + AC$$

$\therefore$ This works by creating a product term that stays true even when B changes. So during the transition from $BC \rightarrow AB'$ when B change $1 \rightarrow 0$, AC stays true, causing the output to stay true and not dip to 0